Jiuyi (Joey) Xu

Updated November, 3rd, 2026

Colorado School of Mines
1500 Illinois St., Golden, CO 80401

jiuyixjiuyi@gmail.com
+1 213-551-6071

EDUCATION

Ph.D. Robotics, Colorado School of Mines, Golden, CO — 08/2024 - Present
- Research Interest: **Vision-Language-Action (VLA) Models, Efficient AI**
- Advisor: Dr. Yangming Shi

M.S. Computer Science, University of Southern California, Los Angeles, CA — 08/2022 - 05/2024

B.E. Software Engineering, Dalian University of Technology, Dalian, China — 09/2017 - 06/2021

RESEARCH EXPERIENCE

Yangming Research Group, Colorado School of Mines — **Golden, CO**
Graduate Research Assistant, Advisor: Dr. Yangming Shi — 08/2024 - Present
- Conduct research in efficient vision-language-action models.
- Conduct research in the application of generative AI across the field of civil engineering.

Geospatial Terrain Research Team, USC Institute for Creative Technologies — **Los Angeles, CA**
Research Assistant Student Intern, Advisor: Dr. Meida Chen — 06/2023 - 06/2024
- Conducted research in 3D semantic segmentation and 3D photogrammetric point cloud data processing:
  a) **Automated Annotation in Indoor/Outdoor Environment**
  Developed a framework to annotate both source and synthetic images automatically via Open-Vocabulary Object Detection (OVOD) and semantic segmentation to solve the limited availability of publicly accessible datasets in civil and construction environments.
  b) **Tripping Hazard Detection in Construction Site**
  Proposed a framework to detect tripping hazards at construction sites using OVOD models (Grounding DINO) and the Segment Anything Model (SAM), highlighting the superior generalization ability of OVOD in civil engineering scenarios compared to traditional methods.
  c) **2D-3D Open-Vocabulary Semantic Segmentation**
  Implemented Open-Vocabulary Semantic Segmentation (OVSS) based on OVOD and k-means clustering algorithm in 3D photogrammetric cloud point data.

WORK EXPERIENCE

Prime Robotics — **Denver, CO**
Robotics Intern, Manager: Ryan Austin — 05/2026 - 08/2026
- Developed, optimized and maintained a production-grade multi-agent path planning system supporting 10+ agents in real-time environments.
- Trained and integrated large-scale AI models to continuously optimize planning accuracy and system throughput.

PUBLICATIONS

- **Xu, J.**, Hu, X., Chen, M., Wang, S., Ye, Y., & Shi, Y. (2026). *Quantize by Doing: Task-Guided Mixed-Precision Quantization of Vision-Language-Action Models*. ArXiv.org. https://arxiv.org/abs/2610.38661
- **Xu, J.**, Chen, M., Ye, Y., & Shi, Y. (2026). *CuraEval: Evaluating Demonstration Curation on Mixed-Skill Robot Data*. "Oops, I Erred" Workshop at the 2026 Conference on Robot Learning (CoRL). https://arxiv.org/abs/2610.38221
- **Xu, J.**, Hu, X., Chen, M., Gao, P., Ye, Y., & Shi, Y. (2026). *World Models Dream of Success: Diagnosing and Repairing Failure Insensitivity in Robot World Models*. ArXiv.org. https://arxiv.org/abs/2610.09134

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- **Xu, J.**, Guo, J., Chen, M., Du, J., & Shi, Y. (2026). *Predict Before You Deploy: Offline Prediction of Quantization-Induced Task Degradation for World Action Models*. ArXiv.org. https://arxiv.org/abs/2609.19441
- **Xu, J.**, Jin, Q., Chen, M., Wang, S., Sui, Y., & Shi, Y. (2026). *VLAQuantBench: Closed-Loop Evaluation of Post-Training Quantization for Vision-Language-Action Models*. ArXiv.org. https://arxiv.org/abs/2609.25376
- **Xu, J.**, Qian, C., Tu, Z., Sui, Y., & Shi, Y. (2026). *FastVLA: Task-Conditioned and Temporally Adaptive Visual Token Compression for Efficient Vision-Language-Action Models*. 2027 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV). https://arxiv.org/abs/2610.22146
- **Xu, J.**, Chen, M., & Shi, Y. (2026). *EgoGen: Generating Egocentric Images on Construction Sites Using Diffusion Model*. Construction Research Congress 2026, 63–72. https://doi.org/10.1061/9780784486962.007
- Vahidi, M., **Xu, J.**, Shi, Y., & Liu, K. (2026). *Domain Knowledge-Infused Prompting for Deep and Full-Sentence Information Extraction from Bridge Inspection Reports*. Journal of Computing in Civil Engineering.
- Yu, H., **Xu, J.**, Shi, Y., & Ye, Y. (2026). *Principal Variate Selection Approach for LSTM Building Load Forecasting*. Proceedings of SimBuild 2026: 12th National Conference of IBPSA-USA, 327–338. https://doi.org/10.26868/30680611.2026.1314
- **Xu, J.**, Kim, R., Ye, Y., & Shi, Y. (2026). *An autoregressive-based framework for simulating stochastic occupant behavior in residential buildings*. Building Simulation. Vol. 19. No. 2. Tsinghua University Press, 2026. https://link.springer.com/article/10.1007/s12273-026-1406-3
- **Xu, J.**, Chen, M., & Shi, Y. (2025). *GenVision: Enhancing Construction Safety Monitoring with Synthetic Image Generation*. 2025 Winter Simulation Conference (WSC). IEEE, 2025. https://ieeexplore.ieee.org/abstract/document/11338881
- **Xu, J.**, Jin, Q., Chen, M., Feng, A., Sui, Y., & Shi, Y. (2025). *LowDiff: Efficient Diffusion Sampling with Low-Resolution Condition*. ArXiv.org. https://arxiv.org/pdf/2509.15342
- Chen, M., Leal, L., Hu, Y., Liu, R., Xiong, B., Feng, A., **Xu, J.**, & Shi, Y. (2025). *IDU: Incremental Dynamic Update of Existing 3D Virtual Environments with New Imagery Data*. 2025 Interservice/Industry Training, Simulation and Education Conference (I/ITSEC). https://arxiv.org/abs/2508.17579
- Liu, Z., **Xu, J.**, Wun, C., Chen, M., Zou, Z., & Shi, Y. (2025). *Egocentric Camera-Based Method for Detecting Static Hazardous Objects on Construction Sites*. Automation in Construction 172 (2025): 106048. https://www.sciencedirect.com/science/article/pii/S0926580525000883
- **Xu, J.**, Chen, M., Feng, A., Yu, Z., & Shi, Y. (2024). *Open-Vocabulary High-Resolution 3D (OVHR3D) Data Segmentation and Annotation Framework*. 2024 Interservice/Industry Training, Simulation and Education Conference (I/ITSEC). https://arxiv.org/abs/2412.06268
- **Xu, J.**, Sanni, T., Liu, Z., Yang, Y., Lee, J., Song, W., & Shi, Y. (2024). *To Shelter or Not To Shelter: Exploring the Influence of Different Modalities in Virtual Reality on Individuals' Tornado Mitigation Behaviors*. ArXiv.org. https://arxiv.org/abs/2409.09205
- Chen, M., Lal, D., Yu, Z., **Xu, J**., Feng, A., You, S., Nurunnabi, A., & Shi, Y. (2024). *Large-Scale 3D Terrain Reconstruction Using 3D Gaussian Splatting for Visualization and Simulation*. The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, XLVIII-2-2024, 49–54. https://doi.org/10.5194/isprs-archives-xlviii-2-2024-49-2024
- Rahman, M. A., **Xu, J.**, Liu, Z., Chen, M., Zhu, R., Liu, Y., Shi, Y., & Feng, A. (2024). *An Open-Vocabulary Framework for Efficient 2D and 3D Visual Data Annotation in Indoor and Outdoor Built Environments*. 2024 ASCE International Conference on Computing in Civil Engineering (i3CE).
- Liu, Z., **Xu, J.**, Suen, W. K., Chen, M., Zou, Z., Feng, A., & Shi, Y. (2024). *Detecting without Training: An Open-Vocabulary Object Detection Method for Identifying Hazardous Objects on Construction Sites*. 2024 ASCE International Conference on Computing in Civil Engineering (i3CE).

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CONFERENCE PRESENTATIONS

- "EgoGen: Generating Egocentric Images on Construction Sites Using Diffusion Model", *CRC 2026*, San Antonio Marriott Rivercenter on River Walk, San Antonio, TX, March 18th – 21st, 2026
- "GenVision: Enhancing Construction Safety Monitoring with Synthetic Image Generation", *WSC 2025*, Sheraton Grand Seattle, Seattle, WA, December 7th – 10th, 2025
- "An Open Vocabulary Framework for Efficient 2D-3D Visual Data Annotation in Indoor and Outdoor Built Environments", *i3CE 2024*, Carnegie Mellon University, Pittsburgh, PA, July 28th – 31st, 2024
- "A Hybrid 2D-3D Zero-shot Framework for Photogrammetric Data Segmentation", *Geo Week 2024*, Colorado Convention Center, Denver, CO, February 11th – 13th, 2024
- "USC-ICT Semantic Terrain Points Labeling System Demo", *I/ITSEC 2023*, West Concourse of the Orange County Convention Center, Orlando, FL, November 27th – December 1st, 2023

PEER REVIEW EXPERIENCE

- Journals: Journal of Computing in Civil Engineering, IEEE Transactions on Neural Networks and Learning Systems, International Journal of Data Science and Analytics
- Conferences: NeurIPS 2026, AISTATS 2026, AISTATS 2027, ICASSP 2027

AWARDS

- NVIDIA Academic Grant (2026)
- American Society of Civil Engineers (ASCE) Best Data Award (2025)
- American Society of Civil Engineers (ASCE) Outstanding Reviewer (2025)